GUNFIRE

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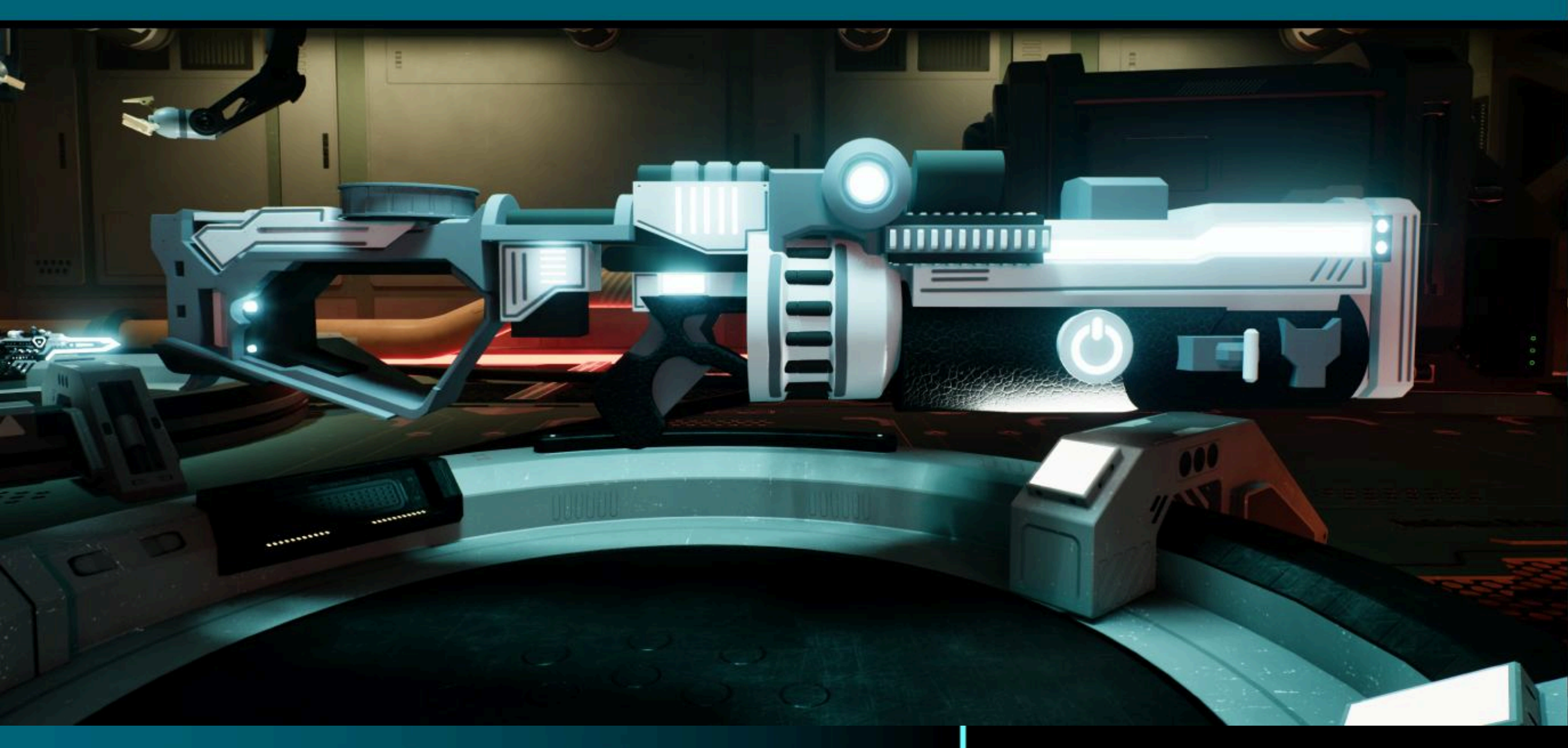
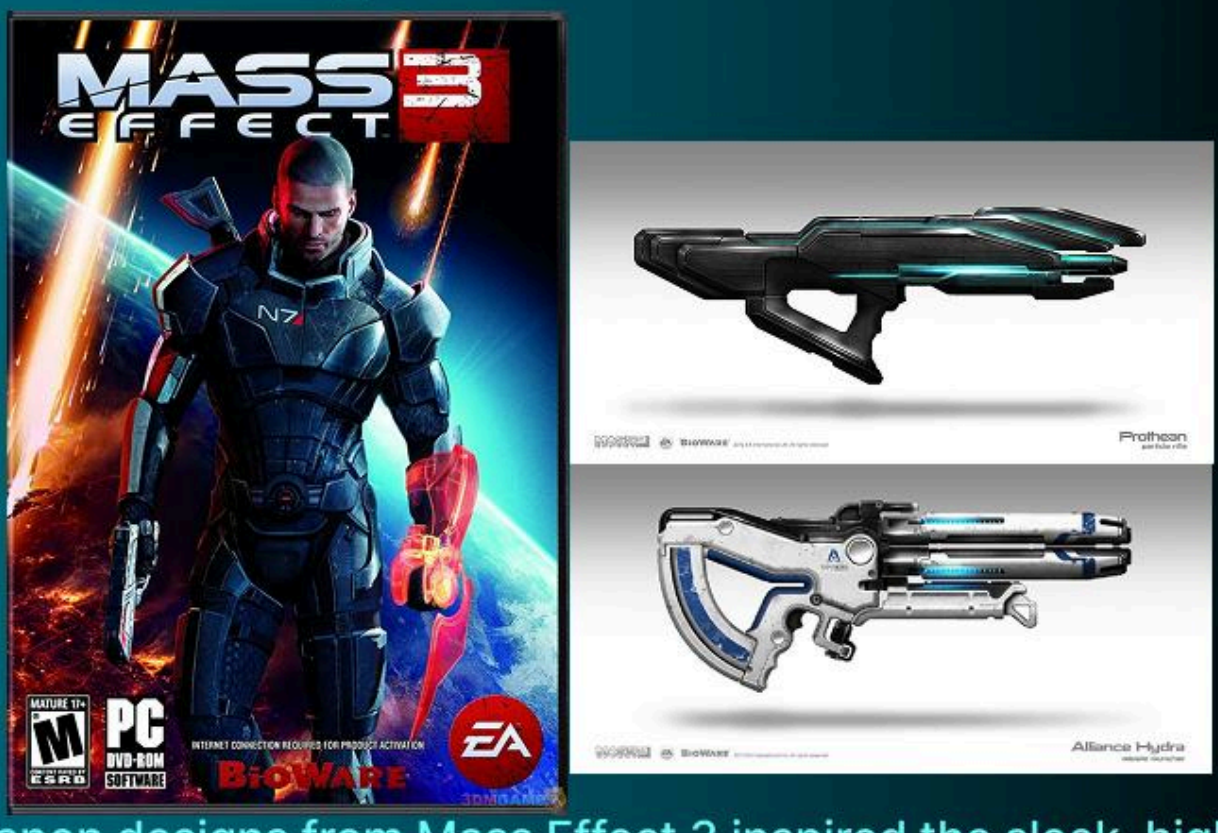
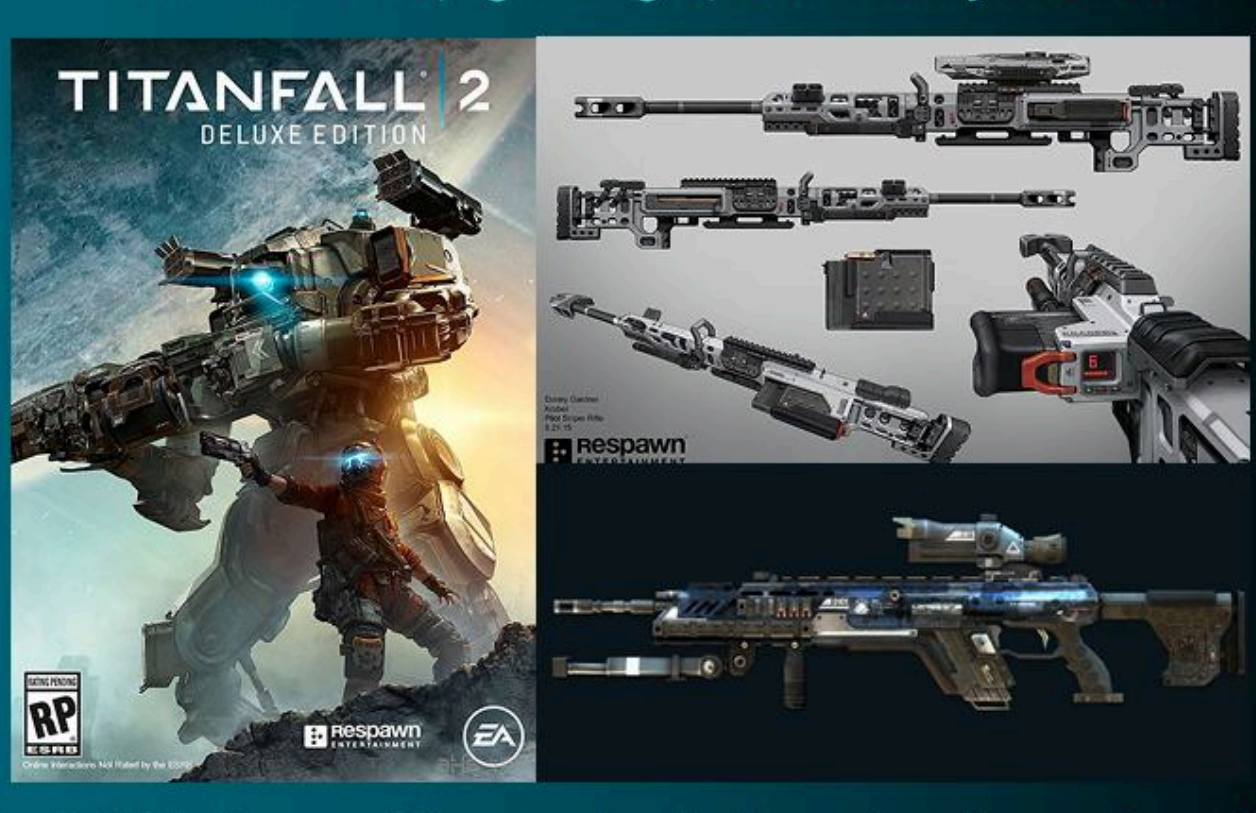
The GUNFIRE Weapon Design Project explores how visual language, material definition, and functional storytelling can work together to create compelling sci-fi weapon assets for modern games. Across four unique firearms, the project demonstrates a complete artistic workflow—from concept exploration and modular design to modeling, texturing, and real-time presentation. Each weapon carries its own identity and gameplay purpose, requiring a careful balance between aesthetics and functionality. Through the use of emissive elements, mechanical silhouettes, and readable forms, the designs enhance player feedback while supporting immersion within a futuristic combat environment.



Inspiration



The weapon designs from Mass Effect 3 inspired the sleek, high-tech aesthetic found in this project. Its clean silhouettes, glowing energy components, and advanced materials helped define the futuristic tone of my weapons. The balance between elegance and cutting-edge technology in Mass Effect guided my approach to creating weapons that feel refined, lightweight, and visually distinct.



Titanfall 2 influenced the mechanical construction and modular complexity of my designs. The game's grounded sci-fi style—featuring exposed hardware, functional details, and believable engineering—helped shape the structural logic behind my weapons. Its emphasis on practicality and mechanical readability encouraged me to incorporate visible components and layered forms to enhance authenticity.

Overview

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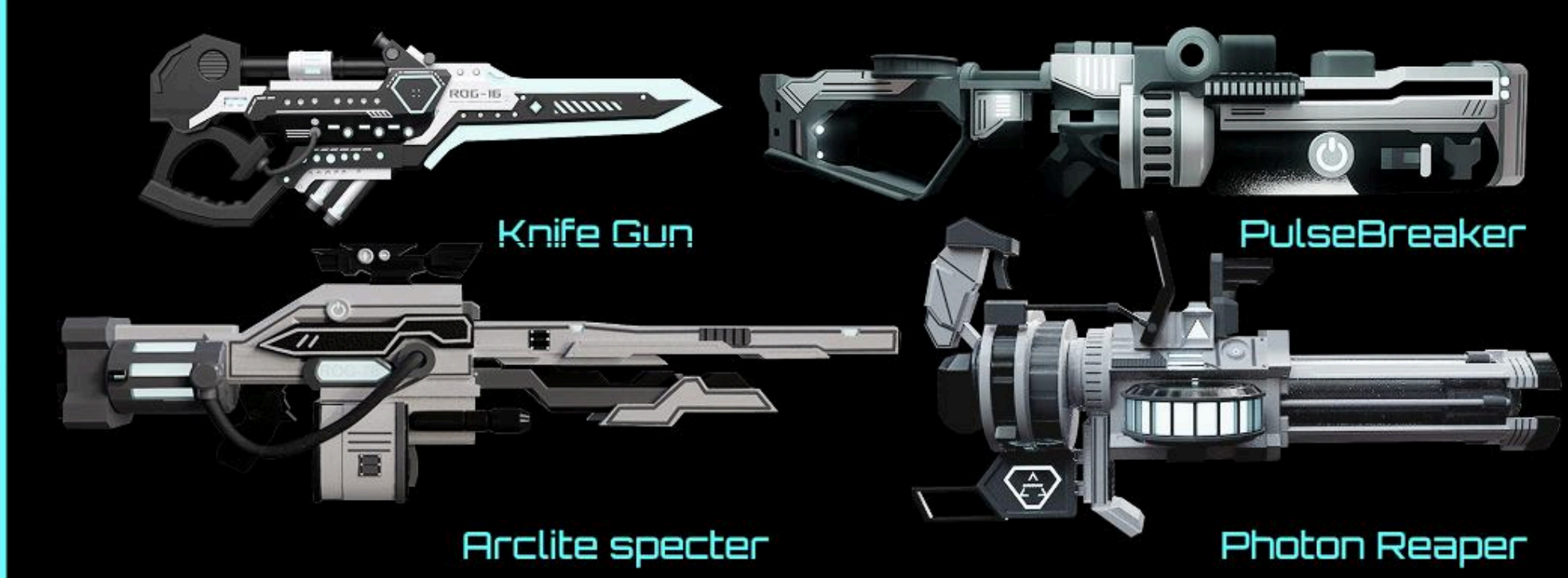
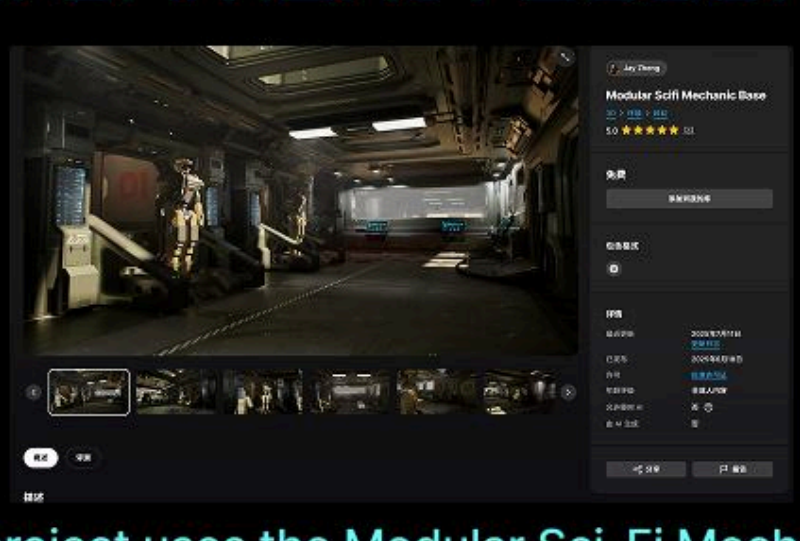


Environment resources:



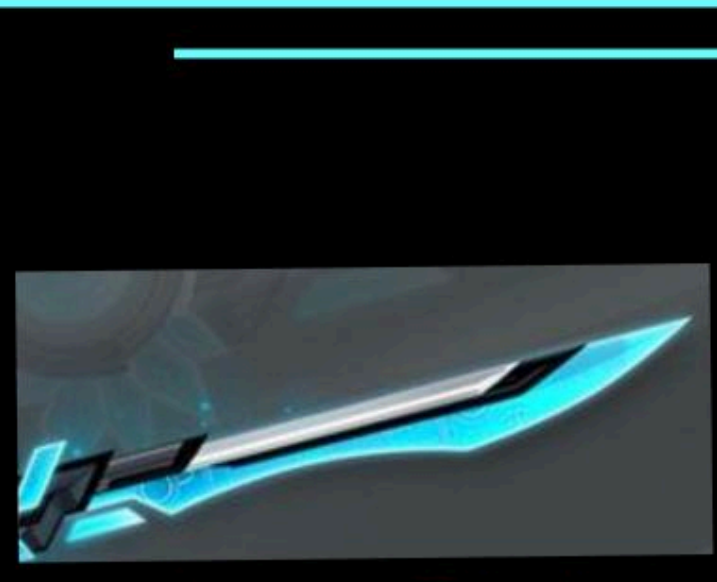
This project uses the Modular Sci-Fi Mechanic Base by Jay Zheng to present the weapons in a believable futuristic environment. Its clean lighting and detailed mechanical design enhance the realism and visual impact of the models.

Knife Gun △

The Knife Gun combines a ranged firearm with a built-in energy blade, designed to support both mid-range combat and close-quarters encounters. The weapon emphasizes speed, precision, and versatility, allowing the user to seamlessly switch between shooting and melee attacks.

Heat Exhaust Unit

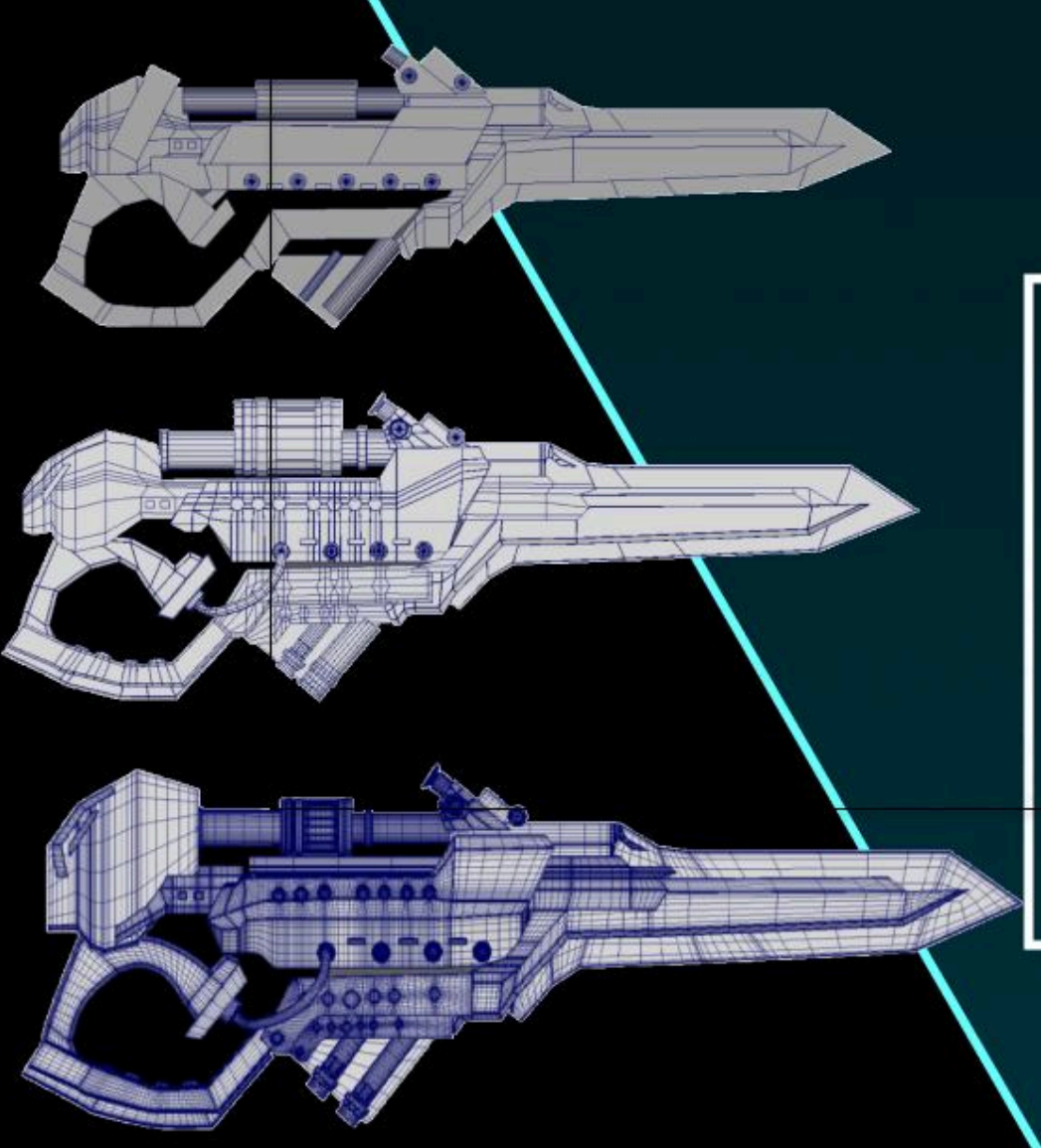
This section highlights the weapon's front energy-vent system, featuring exposed coils and glowing heat elements. These components add functional detail and reinforce the sci-fi power mechanism of the Knife Gun.

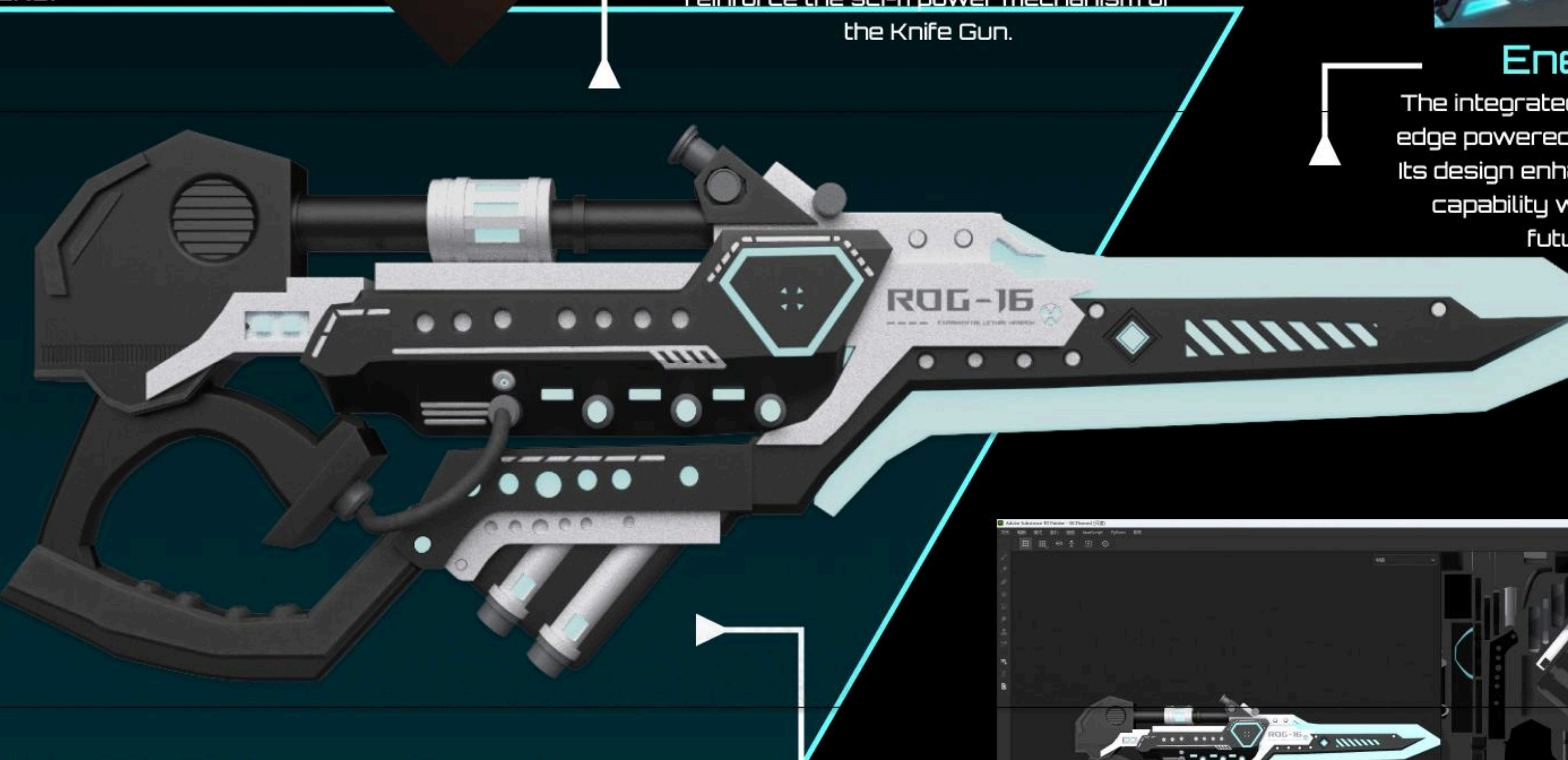


Energy Blade

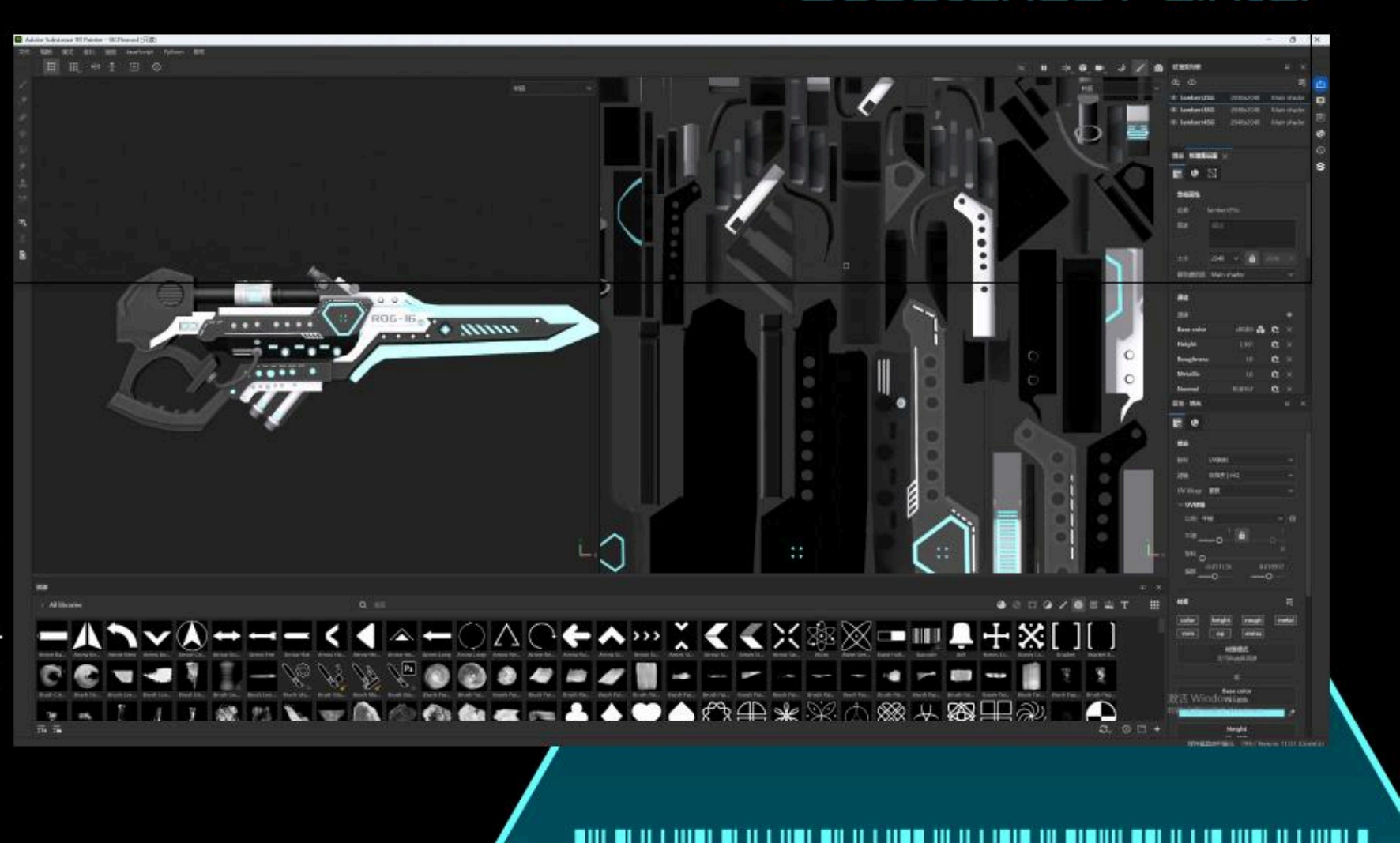
The integrated blade features a glowing edge powered by an internal energy cell. Its design enhances the weapon's melee capability while maintaining a sleek futuristic aesthetic.

MAYA Models





-Substance Painter



Grip & Trigger Section

The grip and trigger assembly are shaped for ergonomic handling, ensuring stability during both firing and melee strikes. Additional mechanical details emphasize functionality.

Energy Cell

A removable energy cell powers both the firing mechanism and the blade. Its placement and design highlight the weapon's dual-system operation.

Knife Gun

PulseBreaker ▢

The PulseBreaker is a mid-range energy rifle built around a high-output pulse system, delivering fast and concentrated shots. Its reinforced frame, detachable energy cell, and ventilated pulse chamber create a clean yet functional sci-fi weapon silhouette.

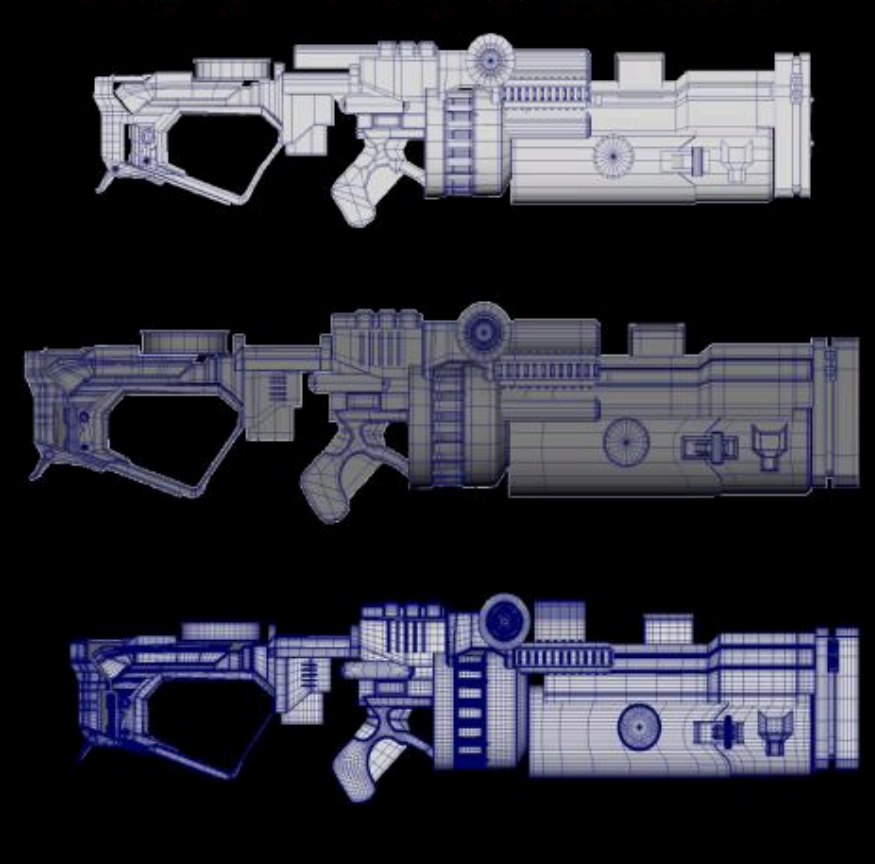
Stock

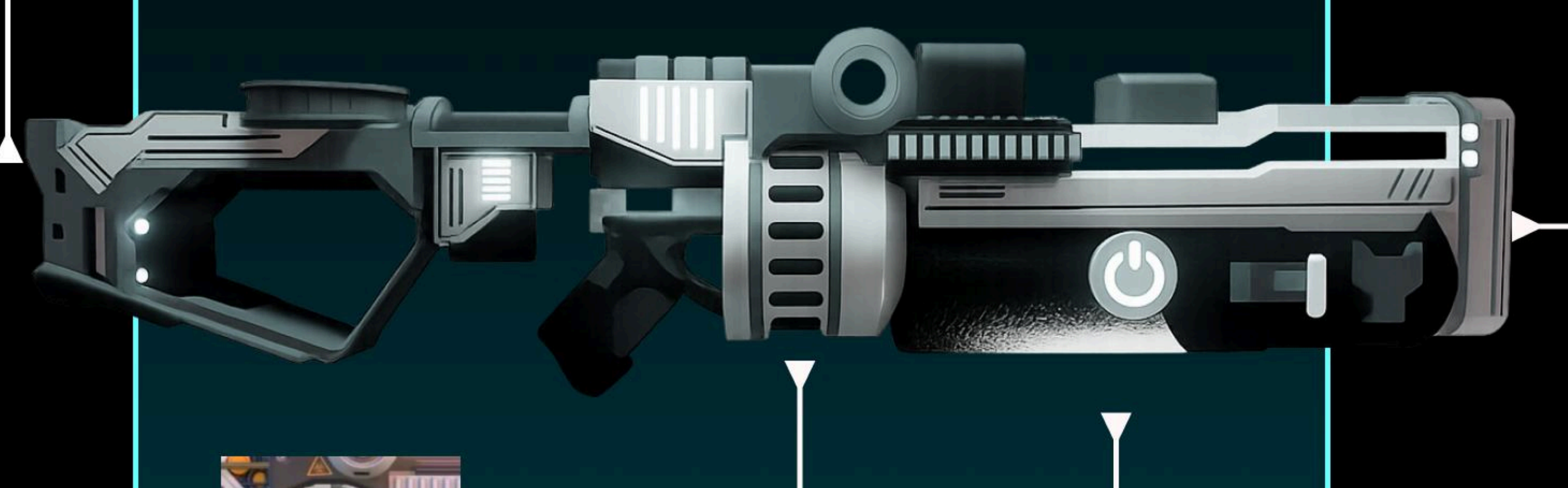
The rear stock is designed for stability and recoil control, featuring layered mechanical plates that reinforce the weapon's structure. Its angular form adds both visual weight and functional support during sustained firing.

Muzzle

The muzzle is engineered to focus and stabilize the pulse discharge, shaping each shot into a concentrated energy beam. Its metallic housing and illuminated core reinforce the weapon's advanced technology.

MAYA Models





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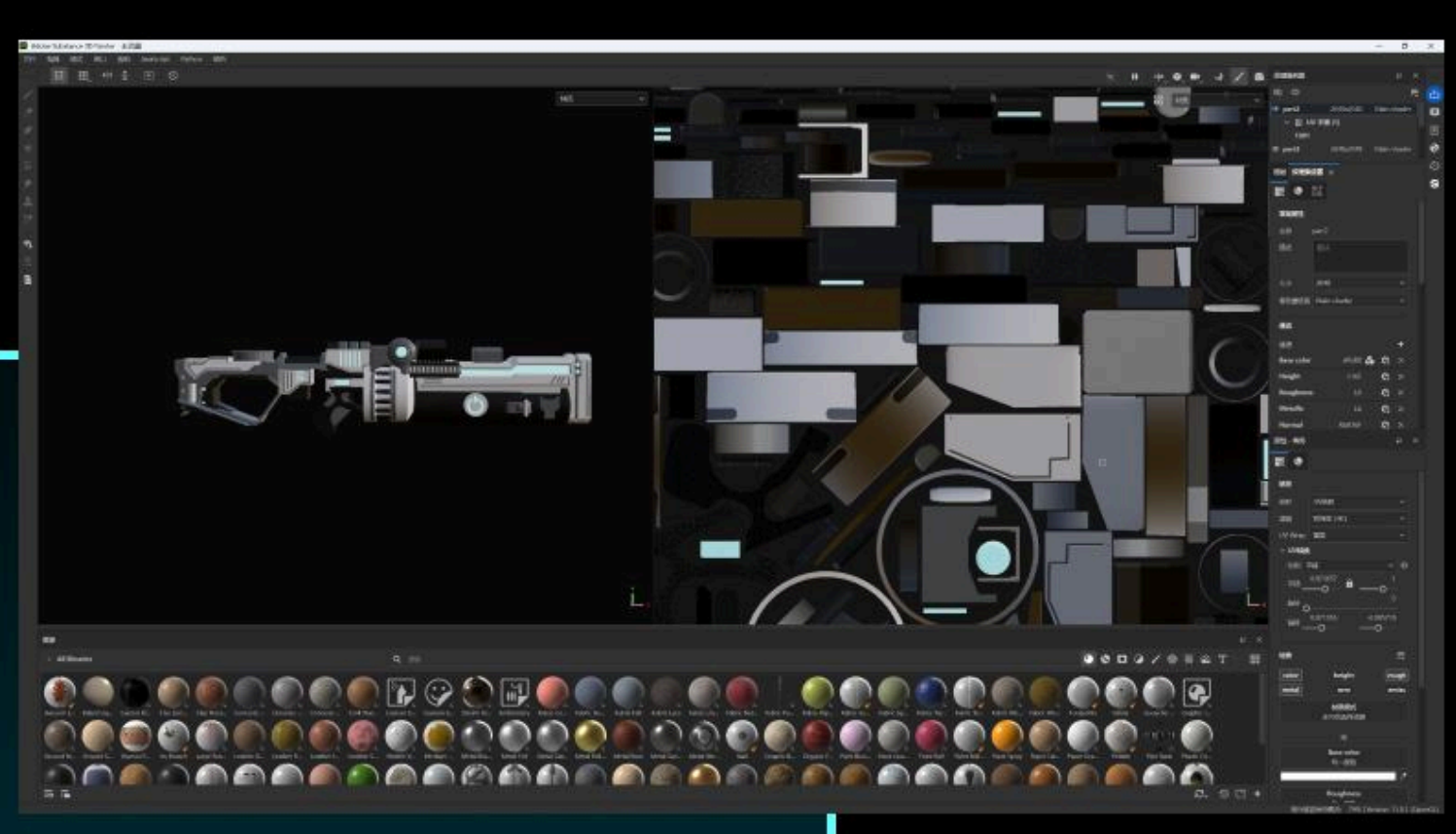


Magazine

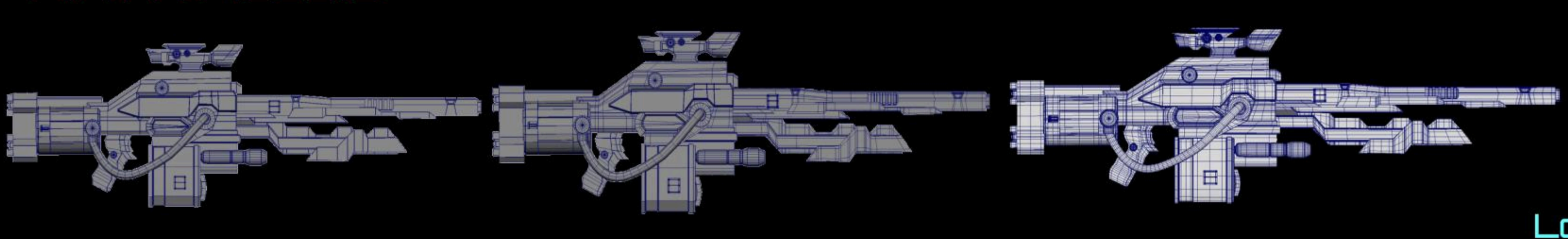
The cylindrical magazine stores high-density pulse cartridges, allowing rapid energy delivery during sustained fire. Its exposed rounds and reinforced housing emphasize the weapon's power-focused design.

Energy Output Module

The energy output module regulates and channels the weapon's pulse power, ensuring stable discharge during each shot. Its illuminated indicators and reinforced casing highlight the high-tech complexity of the system.

PulseBreaker

MAYA Models



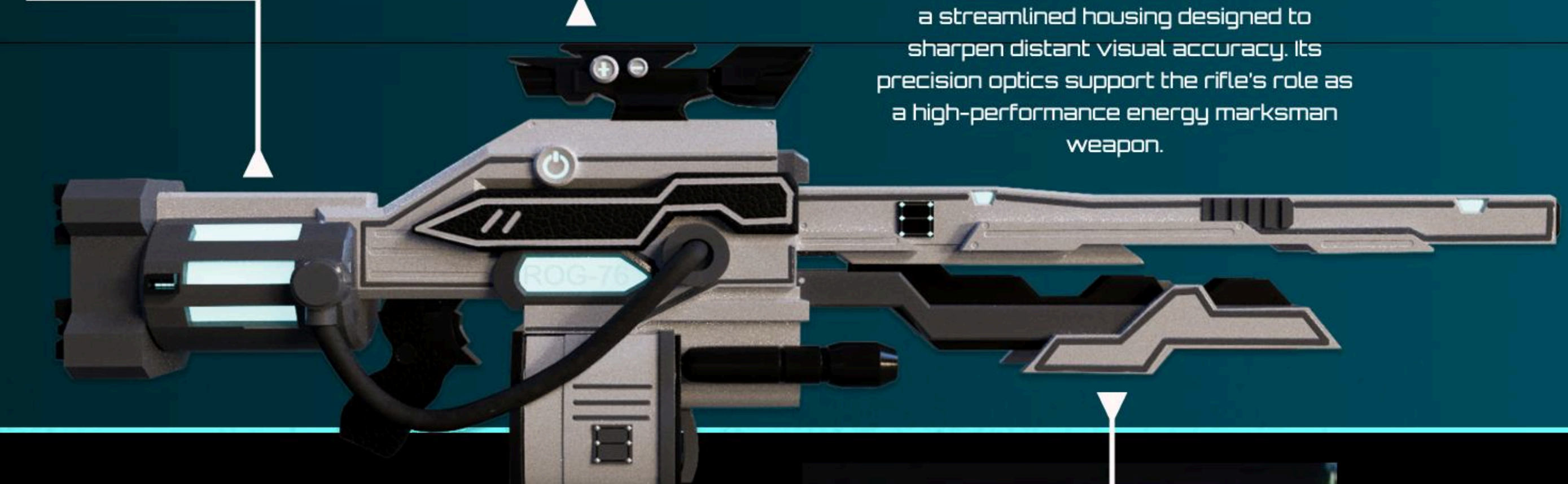
Arclite Specter

Long-Range Targeting Scope

The long-range targeting scope features a streamlined housing designed to sharpen distant visual accuracy. Its precision optics support the rifle's role as a high-performance energy marksman weapon.

Rear Energy Housing

The rear housing integrates structural support with an embedded energy output module. Reinforced plating and illuminated vents stabilize the rifle while regulating the flow of arclight energy during each shot.



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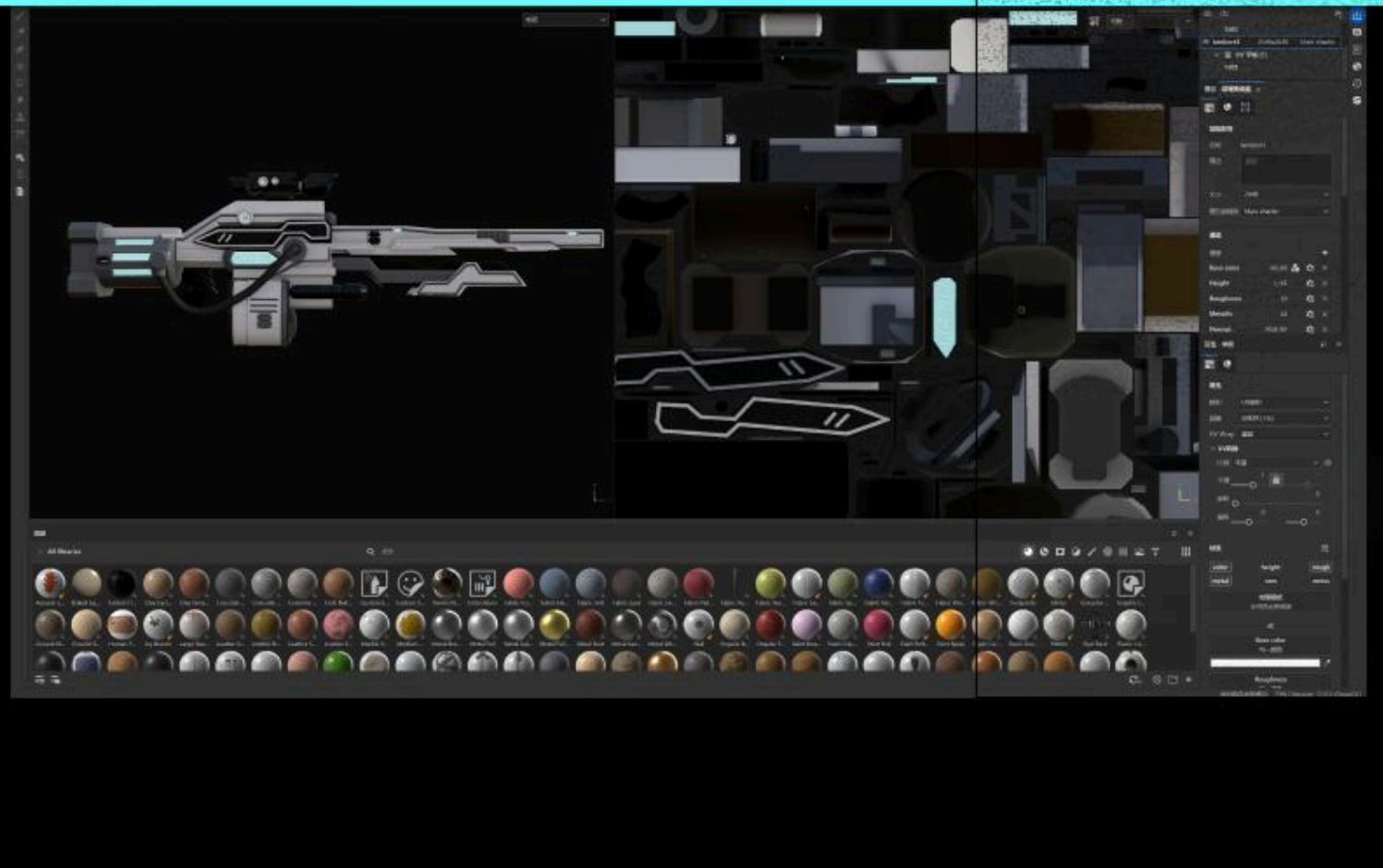
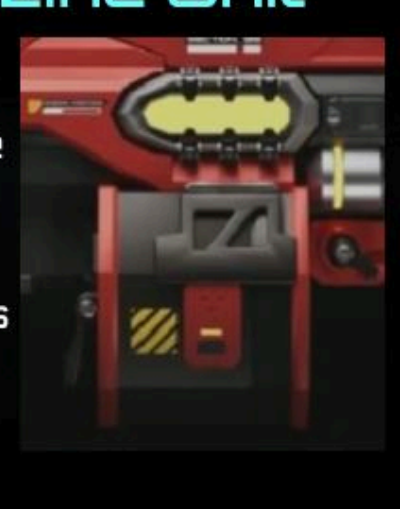


Energy Display & Magazine Unit

The energy display and magazine unit provides real-time power readouts while supplying the rifle with arclight ammunition. Its illuminated gauge and reinforced housing highlight the weapon's high-capacity energy system.

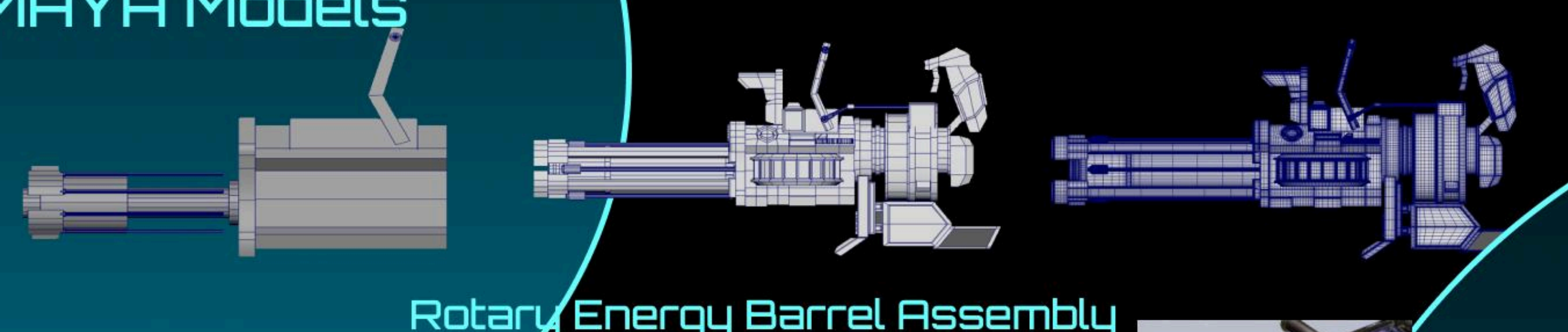


Barrel Assembly & Underframe Rail

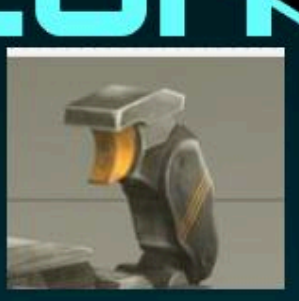
The extended barrel assembly enhances long-range accuracy while maintaining a clean sci-fi profile. The underframe rail supports modular attachments, adding flexibility to the rifle's tactical configuration.

Arclite Specter

MAYA Models



○ Photon Reaper

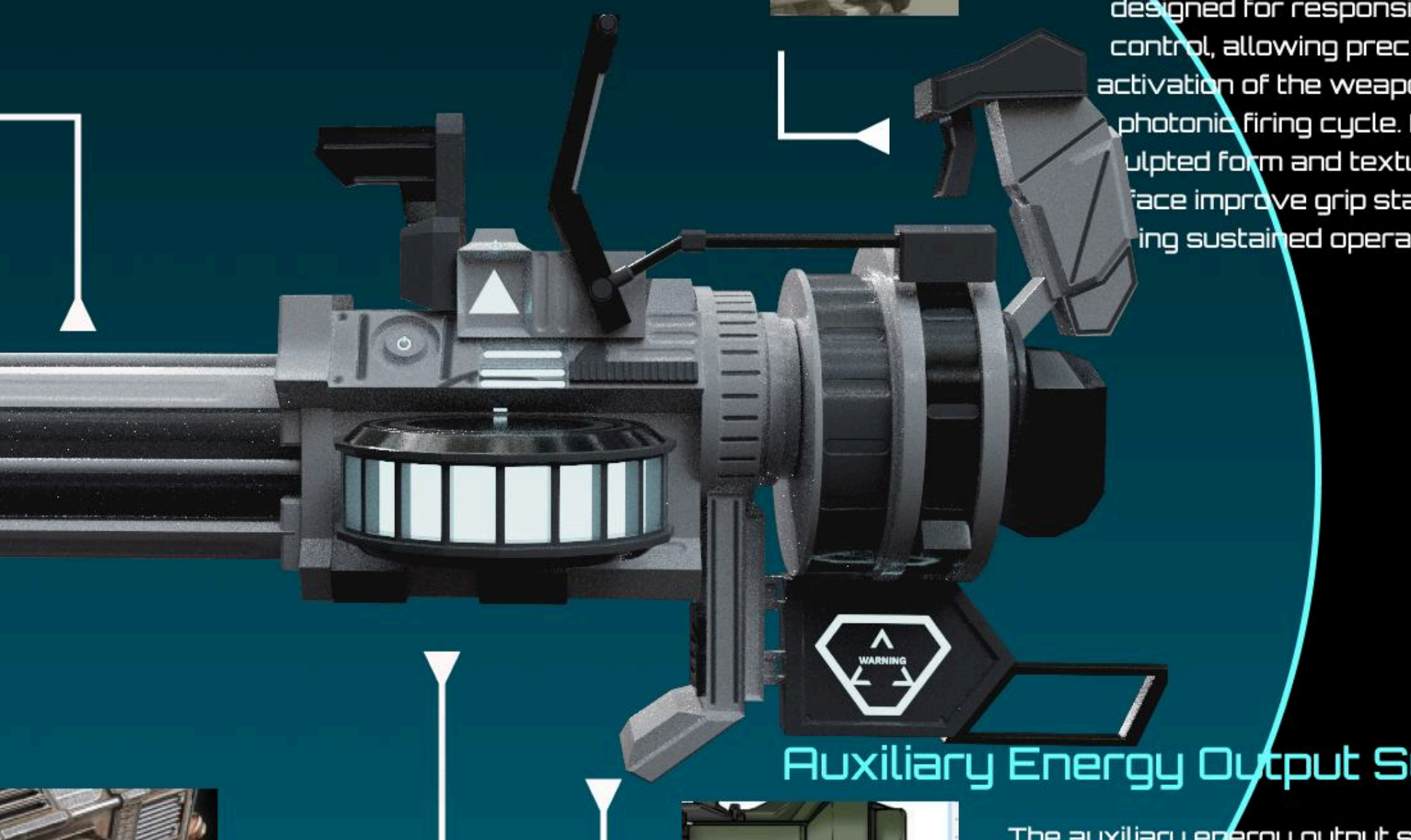


Ergo-Tactile Trigger

The ergo-tactile trigger is designed for responsive control, allowing precise activation of the weapon's photonic firing cycle. Its sculpted form and textured surface improve grip stability during sustained operation.

Rotary Energy Barrel Assembly

The rotary energy barrel assembly channels stabilized photonic currents through multiple spinning conduits, enabling continuous high-output fire. Its transparent housing and exposed emitters highlight the weapon's advanced energy-cycling system.



Photon Convergence Array

The Photon Convergence Array compresses and focuses photonic energy into a stable, high-intensity beam before firing. Its radial emitter plates and illuminated channels emphasize the weapon's advanced energy-focusing technology.

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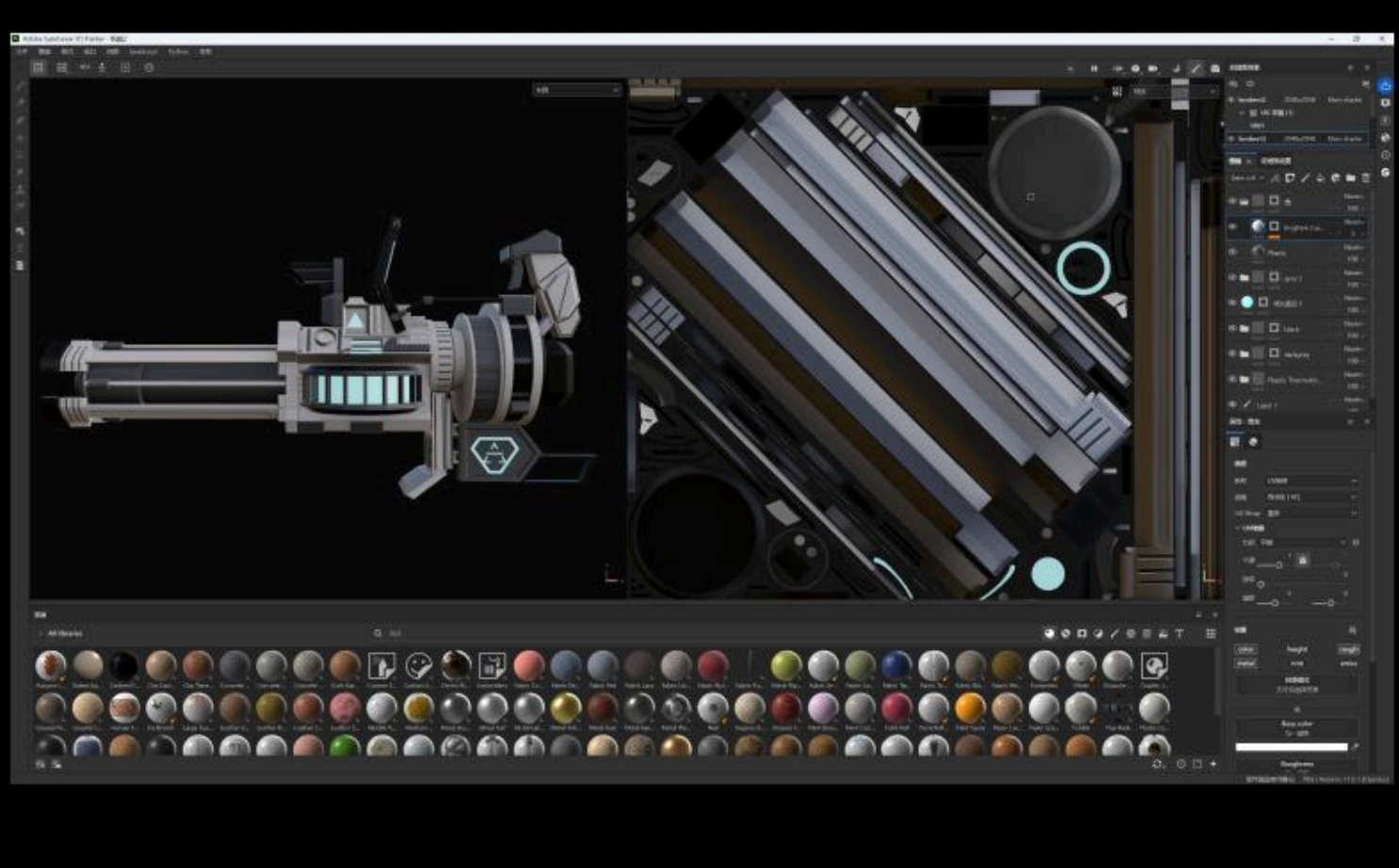

Photon Reservoir Drum

The photon reservoir drum stores condensed photonic fuel and feeds it into the weapon's rotary firing system. Its reinforced housing and vented ring design support prolonged high-energy output.

Auxiliary Energy Output System

The auxiliary energy output system channels additional photonic power into the weapon's firing mechanism, stabilizing its high-output rotation cycle. Its armored housing and structural supports ensure reliable performance under heavy operational loads.

Photon Reaper